



Sequences and Series - Worksheet 1

Chapter 4: Sequences and Series. Answer all questions. Show your working clearly. Total: 126 marks. Time: 2 hours 40 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are six sections, one for each topic in Chapter 4. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- For every question, first decide: *arithmetic or geometric?* and *a term or a sum?*
- Give answers in exact form unless the question says otherwise.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Arithmetic sequences and series	21	
B Geometric sequences and series, sum to infinity	20	
C Sigma notation	18	
D Recurrence relations	19	
E Binomial expansion (positive integer powers)	20	
F Binomial expansion (negative and fractional powers)	24	
Roar-level challenge	4	
Total	126	

Section A Arithmetic sequences and series

21 marks

1. An arithmetic sequence has first term 7 and common difference 4. [2]
Find the 15th term of the sequence.

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2. The 3rd term of an arithmetic sequence is 11 and the 8th term is 31.
Find the first term and the common difference.

[3]

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3. Find the sum of the first 25 terms of the arithmetic series

[3]

$$4 + 9 + 14 + 19 + \dots$$

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4. Find the sum of all the multiples of 6 between 50 and 250.

[4]

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5. Priya starts a savings plan. She saves £100 in the first month, and each month after that she saves £15 more than in the previous month.

(a) Find the amount Priya saves in the 12th month.

[2]

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(b) Find the total amount Priya saves in the first 2 years.

[3]

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(c) Priya's target is to have saved a total of more than £10 000.

[4]

Show that the number of months, n , needed satisfies

$$3n^2 + 37n - 4000 > 0,$$

and hence find the number of months it takes Priya to reach her target.

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(Total for Question 5 is 9 marks)



Section B Geometric sequences and series, sum to infinity

20 marks

6. A geometric sequence has first term 5 and common ratio 3. [2]
Find the 6th term of the sequence.

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7. A geometric series has first term 64 and common ratio $\frac{1}{2}$. [3]
(a) Find the sum of the first 8 terms.
(b) Find the sum to infinity of the series.

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8. The 2nd term of a geometric sequence is 12 and the 5th term is -96 . [3]
Find the common ratio and the first term.

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9. The sum to infinity of a geometric series is four times the first term of the series.

[4]

(a) Find the common ratio.

(b) Given that the third term of the series is 18, find the first term.

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10. The second term of a geometric series is 48 and the sum to infinity of the series is 256.

(a) Show that $16r^2 - 16r + 3 = 0$, where r is the common ratio of the series. [4]

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(b) Hence find the two possible values of the first term of the series. [2]

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(c) For the series with the **larger** common ratio, find the difference between the sum to infinity and the sum of the first 5 terms. [2]

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(Total for Question 10 is 8 marks)



Section C Sigma notation

18 marks

11. Find the value of

[2]

$$\sum_{r=1}^5 (2r + 1).$$

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12. Find the value of

[3]

$$\sum_{r=1}^{20} (5r - 3).$$

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13. Write the series $4 + 7 + 10 + \dots + 61$ using sigma notation, and find its sum.

[3]

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14. Find the value of

[3]

$$\sum_{r=1}^8 3 \times 2^r.$$

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15. (a) Show that $\sum_{r=1}^n (4r+1) = n(2n+3)$.

[3]

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(b) Hence, or otherwise, find the value of $\sum_{r=11}^{30} (4r+1)$.

[2]

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(c) Given that $\sum_{r=1}^k (4r+1) = 377$, find the value of k .

[2]

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(Total for Question 15 is 7 marks)



Section D Recurrence relations

19 marks

16. A sequence is defined by $u_1 = 4$, $u_{n+1} = 3u_n - 2$. [2]

Find u_2 , u_3 and u_4 .

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17. A sequence is defined by $u_1 = 5$, $u_{n+1} = \frac{1}{2}u_n + 3$. [3]

(a) Find u_2 , u_3 and u_4 .

(b) Describe the behaviour of the sequence.

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18. A sequence is defined by $u_1 = 3$, $u_{n+1} = 1 - \frac{1}{u_n}$. [3]

(a) Find u_2 , u_3 and u_4 , and state the period of the sequence.

(b) Hence find u_{100} .

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19. A sequence is defined by $u_1 = 2$, $u_{n+1} = ku_n + 3$, where k is a constant. [4]
Given that $u_3 = 17$, find the possible values of k .

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20. A sequence $a_1, a_2, a_3, \dots$ is defined by

$$a_1 = 5, \quad a_{n+1} = 2a_n - c,$$

where c is a constant.

- (a) Find an expression for a_3 in terms of c . [2]

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- (b) Given that $\sum_{r=1}^3 a_r = 11$, find the value of c . [3]

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- (c) Using your value of c , find a_5 , and state whether the sequence is increasing or decreasing. [2]

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(Total for Question 20 is 7 marks)



Section E Binomial expansion (positive integer powers)

20 marks

21. Expand $(1+x)^4$ fully. [2]

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22. Find the first 3 terms, in ascending powers of x , of the binomial expansion of $(2-x)^6$. [3]

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23. Find the coefficient of x^3 in the binomial expansion of $(3+2x)^7$. [3]

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24. In the binomial expansion of $(1 + kx)^8$, where k is a positive constant, the coefficient of x^2 is 7 times the coefficient of x . [4]

(a) Find the value of k .

(b) Hence find the coefficient of x^3 .

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25. (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of [3]

$$\left(1 - \frac{x}{2}\right)^{10},$$

giving each term in its simplest form.

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- (b) Use your expansion, with a suitable value of x , to find an approximate value of 0.995^{10} . [2]
Give your answer to 5 decimal places.

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- (c) Find the coefficient of x^2 in the expansion of $(3 + 2x)\left(1 - \frac{x}{2}\right)^{10}$. [3]

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(Total for Question 25 is 8 marks)



Section F Binomial expansion (negative and fractional powers)

24 marks

26. Find the binomial expansion of $(1+x)^{-1}$, in ascending powers of x , up to and including the term in x^3 . [3]
State the range of values of x for which the expansion is valid.

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27. Find the binomial expansion of $\sqrt{1+3x}$, in ascending powers of x , up to and including the term in x^3 . [4]
State the range of values of x for which the expansion is valid.

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28. Find the binomial expansion of

[5]

$$\frac{1}{\sqrt{4-x}}, \quad |x| < 4,$$

in ascending powers of x , up to and including the term in x^2 , giving each coefficient as a simplified fraction.

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29. Find the first 4 terms, in ascending powers of x , of the expansion of

[4]

$$\frac{1+2x}{(1-x)^2}, \quad |x| < 1.$$

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30. (a) Find the binomial expansion of

$$\sqrt{1-8x}$$

[4]

in ascending powers of x , up to and including the term in x^3 , simplifying each coefficient.

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(b) State the range of values of x for which the expansion is valid.

[1]

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(c) Show that, when $x = \frac{1}{100}$, $\sqrt{1-8x} = \frac{\sqrt{23}}{5}$.

[3]

Hence use your expansion to find an approximation for $\sqrt{23}$, giving your answer to 5 decimal places.

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(Total for Question 30 is 8 marks)



ROAR-LEVEL CHALLENGE

4 marks

31. Find the coefficient of x^2 in the expansion of $\frac{(1+2x)^5}{1-x}$, $|x| < 1$. [4]

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END OF WORKSHEET

Total: 126 marks